Machine Learning - Undergraduate Level Assessment 11 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. High entropy means that the partitions in classification are
   (a) pure
   (b) not pure
   (c) useful
   (d) useless
2. Which of the following statements about Naive Bayes is incorrect?
   (a) Attributes are equally important.
   (b) Attributes are statistically dependent of one another given the class value.
   (c) Attributes are statistically independent of one another given the class value.
   (d) Attributes can be nominal or numeric
3. Statement 1| RoBERTa pretrains on a corpus that is approximate 10 x larger than the corpus BERT pretrained on. Statement 2| ResNeXts in 2018 usually used tanh activation functions.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True
4. Which of the following is the joint probability of H, U, P, and W described by the given Bayesian Network H -> U <- P <- W? [note: as the product of the conditional probabilities]
   (a) P(H, U, P, W) = P(H) * P(W) * P(P) * P(U)
   (b) P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(W | H, P)
   (c) P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(U | H, P)
   (d) None of the above
5. Which of the following best describes what discriminative approaches try to model? (w are the parameters in the model)
   (a) p(y|x, w)
   (b) p(y, x)
   (c) p(w|x, w)
   (d) None of the above

True / False (10 marks)

Answer True or False

6. True or False: DCGANs utilize self-attention mechanisms to enhance training stability.
7. True or False: Highway networks were introduced prior to ResNets and utilize max pooling instead of convolutions.
8. True or False: Dropout randomly removes entire neurons from the network instead of scaling activation values to zero.
9. True or False: The cost of one gradient descent update given the gradient is O(ND), where N is the number of data points and D is the number of features.
10. True or False: Regression is primarily used for prediction and relates inputs to outputs in a quantitative manner.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to extract all the adjacent coordinates of the given coordinate tuple.
12. Write a function to find number of odd elements in the given list using lambda function.

End of Paper